

Atomic Structure 8

Essential Pre-Uni Chemistry D1.8

Subject & topics: Chemistry | Foundations | Atomic Structure **Stage & difficulty:** A Level P2

An ion of nickel is found to have the ground state electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$ in the gas phase.

Give the numerical charge on the ion as an integer. Remember to include the appropriate sign in your answer (as $+N$ or $-N$ and **not** $N\pm$).

Question deck:

STEM SMART Chemistry Week 14

Super-heavy Water

Subject & topics: Chemistry | Foundations | Atomic Structure **Stage & difficulty:** A Level C1

Hydrogen exists as a mixture of three isotopes: normal hydrogen, deuterium (which can be represented by the symbol D), and tritium, T.

The relative isotopic mass of D is 2.0

The relative isotopic mass of T is 3.0

Assuming that molecules of the tritiated water pack as densely as those in normal water, which has a density of 1.00 g cm^{-3} , calculate the density of liquid T_2O . Use the masses given above and any relative atomic masses from the periodic table, as appropriate, and give your answer to 3 significant figures.

Adapted with permission from UCLES, Chemistry STEP 1998, Question 2

Question deck:

STEM SMART Chemistry Week 14

First Configurations

Subject & topics: Chemistry | Foundations | Atomic Structure**Stage & difficulty:** A Level P2**Part A****Unpaired electron**

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has an unpaired electron in its ground-state configuration.

Part B**Incomplete shell, no unpaired electrons**

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has an incomplete shell, but no unpaired electrons in its ground-state configuration.

Part C**Cation with unpaired electron**

Specify the symbol of the element with the lowest atomic number that satisfies the following property: its singly-charged cation has an unpaired electron in its ground-state configuration.

Part D

Full shell configuration $^{2-}$ anion

Specify the symbol of the element with the lowest atomic number that satisfies the following property: its doubly-charged anion has only full shells in its ground-state configuration.

Part E

Cation and anion

Specify the symbol of the element with the lowest atomic number that satisfies the following property: both its singly-charged cation and its singly-charged anion have two unpaired electrons in their ground-state configurations.

Part F

Partially-filled p-orbital

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has a partially-filled p-orbital in its ground-state configuration.

Part G

Fully-filled p-orbital

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has a fully-filled p-orbital in its ground-state configuration.

Part H

Six unpaired electrons

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has six unpaired electrons in its ground-state configuration.

Part I

Fully-filled d-orbital

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has a fully-filled d-orbital in its ground-state configuration.

Part J

Fully-filled d-subshell

Specify the symbol of the element with the lowest atomic number that satisfies the following property: it has a fully-filled d-subshell in its ground-state configuration.

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Question deck:

STEM SMART Chemistry Week 14

Elements Reversal

Subject & topics: Chemistry | Foundations | Atomic Structure **Stage & difficulty:** A Level C1

Although the elements in Mendeleev's table are primarily arranged by atomic mass, this was not the case with tellurium (Te) and iodine (I). Mendeleev realised that the chemical properties of the elements meant that tellurium had to come before iodine, but the atomic masses did not support this order. He marked the mass of tellurium with a question mark to highlight its suspicious value.

The modern value for the relative atomic mass of tellurium is one of the least precise: 127.60 ± 0.03 . The reason for the uncertainty is that naturally occurring tellurium is a mix of 8 different isotopes whose proportions can vary depending on the sample. In contrast, naturally occurring iodine consists of a single isotope – iodine 127 – and so its relative mass is known to a high degree of precision: 126.904472 ± 0.000003 .

Part A

Protons in iodide

State the number of protons in an iodide ion.

Part B

Neutrons in iodide

State the number of neutrons in an iodide ion.

Part C

Electrons in iodide

State the number of electrons in an iodide ion.

Part D

Tellurium-130

The heaviest of the isotopes found in naturally occurring tellurium is tellurium-130 which has a relative mass of 129.906223. Technically, tellurium-130 is very slightly radioactive and if there were none in the naturally occurring element, the relative atomic mass of tellurium would be 126.412449 (which would make it less than iodine).

Calculate the percentage of tellurium-130 present in naturally-occurring tellurium to 4 significant figures.

Adapted with permission from the Cambridge Chemistry Challenge 2019, Question 1

Question deck:

STEM SMART Chemistry Week 14

Ar & Mr and Molecular Formula 2

Essential Pre-Uni Chemistry A2.2

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level P1

Fluorine only occurs naturally in one isotope, ^{19}F , and has a relative atomic mass of 19.0.

Calculate the mass of a fluorine atom in kg. Give your answer to 3 significant figures.

Question deck:

STEM SMART Chemistry Week 14

Mass Spectrometry

with focus on time-of-flight

Subject & topics: Chemistry | Analytical | Mass Spectrometry **Stage & difficulty:** A Level C2

Mass spectrometry is a technique for studying chemical substances. High resolution mass spectrometry allows us to determine relative atomic or molecular masses to a high degree of precision. Some forms of mass spectrometry can also provide structural information which is useful for structure determination.

Part A

Electron ionisation

A mass spectrometer relies on the presence of ions, which it is able to separate by their mass to charge ratio. However, many samples we wish to study are neutral. The instrument therefore needs to use an ionisation technique to form these ions within the sample. We will first consider **electron ionisation (EI)**. This involves firing high energy electrons at the vaporised sample in order to remove electrons.

Which of the following equations is an accurate description of this electron ionisation process:

- ☐ $X(g) \longrightarrow X^{-}(g) + e^{-}$
- ☐ $X(g) + e^{-} \longrightarrow X^{-}(g)$
- ☐ $X^{+}(g) + e^{-} \longrightarrow X(g)$
- ☐ $X(g) + e^{-} \longrightarrow X^{+}(g) + 2e^{-}$

Part B

Fragmentation

Electron ionisation is a hard ionisation technique, which means it often results in fragmentation of the molecular ion formed initially. As a consequence, when applied to large molecules, the mass spectrum often contains many features corresponding to these lighter ions. This can be helpful for structure determination: for a given molecular structure, we can predict likely ways in which the molecular ion might fragment (e.g. through particular bonds breaking) and check whether this is consistent with the spectrum. That said, some of the fragmentations are far more complex than simple bond breaking and involve rearrangements. This can make it difficult to account for the formation of certain ions even if we confidently know the molecular structure already.

A student was thinking about fragmentation in mass spectrometers and hypothesised that if they saw a fragment feature 15 mass units lighter than the molecular ion, they should also see a feature due to an ion with mass 15. However, consulting some real spectra, they found that it is in fact common not to see ion pairs that “complement” each other like this. By carefully considering a single, simple fragmentation process occurring on a molecular ion, suggest why this might be.

Part C

Electrospray ionisation

A softer ionisation technique, resulting in far less fragmentation than electron ionisation, is **electrospray ionisation**. There are a number of variations of this, but in one type, the neutral atom or molecule bonds to a proton (hydrogen ion).

Select the correct equation describing this process for sodium:

- ☐ $\text{Na(g)} + \text{H}^+(\text{g}) \longrightarrow \text{Ne}^+(\text{g})$
- ☐ $\text{Na(g)} + \text{H}^+(\text{g}) \longrightarrow [\text{Na} + \text{H}]^+(\text{g})$
- ☐ $\text{Na(g)} + \text{H}^+(\text{g}) \longrightarrow \text{Mg}^+(\text{g})$
- ☐ $\text{Na(g)} + \text{H}^-(\text{g}) \longrightarrow \text{Ne}^-(\text{g})$
- ☐ $\text{Na(g)} + \text{H}^-(\text{g}) \longrightarrow [\text{Na} + \text{H}]^-(\text{g})$
- ☐ $\text{Na(g)} + \text{H}^-(\text{g}) \longrightarrow \text{Mg}^-(\text{g})$

Part D

Time-of-flight expression

In a time-of-flight (TOF) mass spectrometer, the ions formed are accelerated through a potential so that all singly-charged ions have the same kinetic energy, E . These ions then travel along a flight tube of length d before reaching the detector. The delay is a function of the mass of a given ion, m .

Derive a formula for the time it takes an ion to travel along the flight tube (the time of flight) as a function of E , d and m . Assume that all quantities are provided in SI units, so your formula does not need to take into account any conversion factors.

The following symbols may be useful: E , d , m

Part E

Calculating molecular mass

Based on the above formula, ions of different mass take different amounts of time to travel through the spectrometer. Calculate, in kg, the mass of an ion X^+ that takes 7.55×10^{-6} s to travel through the 2.00 m drift region, assuming it was previously accelerated to a kinetic energy of 5.83×10^{-15} J.

Part F

Converting to atomic mass units

Convert the above mass of X^+ in kg to atomic mass units (amu) in which you more typically express the mass of atoms, molecules and ions in chemistry.

Part G

Electron mass

To obtain the mass of the neutral species, X , we would actually need to add on the mass of an electron. Why is this not a particularly meaningful calculation for the above quantities?

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Question deck:

STEM SMART Chemistry Week 14

Higher-charged Ions

(in mass spectrometry)

Subject & topics: Chemistry | Analytical | Mass Spectrometry **Stage & difficulty:** A Level C1

A small proportion of ions with higher charge can form during the ionisation process in a mass spectrometer. If a given potential accelerates a singly-charged ion to have a kinetic energy of E , a doubly-charged ion will instead acquire a kinetic energy of $2E$, a triply-charged ion a kinetic energy of $3E$, ... and an ion with charge z a kinetic energy of zE . (If you are interested in the physics behind this, you may wish to read more about [Electric fields](#) and complete a [physics question on accelerating charged particles](#).)

Part A

Time of flight

Derive an expression for the time taken to travel through the drift region of length d by an ion with mass m and charge z , if it has a kinetic energy of zE .

The following symbols may be useful: E , d , m , z

Part B

Isotopic clusters

Ions containing atoms with several stable isotopes form characteristic so-called "isotopic clusters". For example, the molecular ion HCl^+ would be expected to give rise to two lines in the mass spectrum due to the two **isotopologues** H^{35}Cl^+ and H^{37}Cl^+ . The lines would be expected to appear in a 3:1 ratio due to this being the ratio of abundances of the respective chlorine isotopes.

At what (integer) m/z value would you expect the lighter of the two ions above to appear?

At what (integer) m/z value would you expect the heavier of the two ions above to appear?

Part C

Doubly positive

Now consider the two possible isotopologues for HCl^{2+} which has formed in the ionisation stage.

At what (integer) m/z value would you expect the lighter of the two ions above to appear?

At what (integer) m/z value would you expect the heavier of the two ions above to appear?

Part D

More generally

If you know that a compound contains one chlorine atom, how can you identify the charge of the ion responsible for a particular ion cluster (assuming the ion contains the chlorine atom present in the original molecule)?

If you know that a compound contains chlorine, you can relatively easily identify the charge of the ion responsible for a particular isotopic cluster containing at least one chlorine atom. Give an expression for how many units apart individual, neighbouring lines in the isotopic cluster (corresponding to the different isotopologues) will appear on the mass spectrum, as a function of the charge on the ion, z .

The following symbols may be useful: z

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Question deck:

STEM SMART Chemistry Week 14

Ar & Mr and Molecular Formula 7

Essential Pre-Uni Chemistry A2.7

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** GCSE C2, A Level P2

Complete combustion of compound N occurs in a stoichiometric ratio of 1 : 6 with oxygen gas. Complete combustion of 4.2 g of compound N produces 13.2 g of carbon dioxide and 5.4 g of water.

Find the molecular formula of N.

Question deck:

STEM SMART Chemistry Week 14

Solutions 1

Essential Pre-Uni Chemistry B5.1

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** GCSE P2, A Level P1

Calculate the concentration in mol dm^{-3} of the following solutions:

Part A

(a)

0.40 g NaOH in 100 ml water

Part B

(b)

7.3 g HCl in 1000 ml water

Part C

(c)

2.5 g H_2SO_4 in 50 ml water

Part D

(d)

15 g FeSO_4 in 500 ml water, to 2 significant figures

Part E

(e)

0.16 g KMnO_4 in 200 ml

Question deck:

STEM SMART Chemistry Week 14

Homeopathy

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level C2

Part A

Arsenous acid

As_2O_3 is moderately soluble in water: one dm^3 of a saturated solution at 25°C contains 20.6 g. When dissolved in water, the oxide reacts to form arsenous acid, H_3AsO_3 .

Balance the equation for the formation of arsenous acid from As_2O_3 , using the smallest possible integer coefficients (do not include state symbols).

Part B

Saturated solution concentration

As_2O_3 is moderately soluble in water: one dm^3 of a saturated solution at 25°C contains 20.6 g. When dissolved in water, the oxide reacts to form arsenous acid, H_3AsO_3 .

Calculate the concentration of the arsenous acid in mol dm^{-3} in the saturated solution, giving your answer to 3 significant figures.

Part C

Mass in glass

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Arsenicum album is often sold as a D30 preparation. Let us assume that the initial stock solution before dilution was the saturated solution containing 20.6 g dm^{-3} of As_2O_3 .

Calculate the mass (in g) of As_2O_3 present in a 100 cm^3 glass of the D30 *Arsenicum album* remedy, giving your answer to 3 significant figures.

Part D

Fatal dose

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Arsenicum album is often sold as a D30 preparation. Let us assume that the initial stock solution before dilution was the saturated solution containing 20.6 g dm^{-3} of As_2O_3 .

Given that 0.1 g of As_2O_3 is usually fatal, calculate the volume (in m^3) of the D30 solution that would be needed for a fatal dose of As_2O_3 , giving your answer to 3 significant figures.

Part E

Volume equivalent

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Arsenicum album is often sold as a D30 preparation. Let us assume that the initial stock solution before dilution was the saturated solution containing 20.6 g dm^{-3} of As_2O_3 .

Given that 0.1 g of As_2O_3 is usually fatal, calculate the volume of the D30 solution that would be needed for a fatal dose of As_2O_3 , expressing your answer as a multiple of the volume of the Earth (approximately $1.08 \times 10^{12} \text{ km}^3$), i.e. how many Earths would fill the same volume, to 2 significant figures.

Part F

Bottles for one atom

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Arsenicum album is often sold as a D30 preparation. Let us assume that the initial stock solution before dilution was the saturated solution containing 20.6 g dm^{-3} of As_2O_3 .

The preparations are usually sold in one ounce bottles ($1 \text{ ounce} = 28 \text{ cm}^3$).

Calculate how many bottles of the D30 solution need to be bought in order, on average, to include one atom of arsenic.

Part G

D2

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Another 'remedy' is *Muriaticum acidum* which is actually diluted hydrochloric acid. This is available in various dilutions up to the extremely 'potent' D2000 preparation.

Assuming the original stock solution before dilution was 1.0 mol dm^{-3} , what is the concentration of HCl in D2?

Part H

D4

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Another 'remedy' is *Muriaticum acidum* which is actually diluted hydrochloric acid. This is available in various dilutions up to the extremely 'potent' D2000 preparation.

Assuming the original stock solution before dilution was 1.0 mol dm^{-3} , what is the concentration of HCl in D4?

Part I

D6

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Another 'remedy' is *Muriaticum acidum* which is actually diluted hydrochloric acid. This is available in various dilutions up to the extremely 'potent' D2000 preparation.

Assuming the original stock solution before dilution was 1.0 mol dm^{-3} , what is the concentration of HCl in D6?

Part J

D8

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Another 'remedy' is *Muriaticum acidum* which is actually diluted hydrochloric acid. This is available in various dilutions up to the extremely 'potent' D2000 preparation.

Assuming the original stock solution before dilution was 1.0 mol dm^{-3} , what is the concentration of HCl in D8?

Part K

D10

In homeopathy, a decimal-scale is often used to specify the dilution of a given sample: D1 (sometimes labelled 1X) means the sample has been diluted 1 part in 10. D2 (or 2X) means the sample has been diluted 1 in 10, then 1 part of that solution has been further diluted 1 in 10 again to give a 1 in 100 dilution. A D6 (or 6X) solution has repeated this process six times to give a final dilution of 1 in 10^6 .

Another 'remedy' is *Muriaticum acidum* which is actually diluted hydrochloric acid. This is available in various dilutions up to the extremely 'potent' D2000 preparation.

Assuming the original stock solution before dilution was 1.0 mol dm^{-3} , what is the concentration of HCl in D10?

Adapted with permission from the Cambridge Chemistry Challenge 2012, Question 1

Question deck:

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